ALC 2025/2026

2nd Project – Nurse Scheduling Problem using SMT

07/10/2025, version 1.0

Overview

The 2nd ALC project is to develop a software tool for solving the Nurse Scheduling Problem (NSP) problem. In order to solve this problem, students must use a Satisfiability Modulo Theory (SMT) solver.

Context

Consider an healthcare facility such as a clinic or an hospital. It is necessary to make a schedule of the nurses operating in the facility to cover all work shifts that span 24 hours per day.

Building the schedule is not trivial, since there are many constraints from both management, nurses and mandatory safety requirements. For instance, each shift requires a minimum number of nurses to be assigned. On the other hand, nurses have unavailability periods where they cannot be called to work, different number of daily limits of hours to work, among other issues.

Your goal is to develop a tool to solve an instance of the Nurse Scheduling Problem (NSP) using a SMT solver. The tool must output a satisfiable schedule to the problem where the overall value of the nurse's preferences is optimized.

Problem Specification

Let D define the number of days and let S define the number of shifts in each day. You can assume that each shift has the same duration. For instance, if $S = 4$, then each day is split into 4 shifts of 6 hours each. Each shift has a minimum number of required nurses to be able to operate properly. Let $R_{s,d}$ define the required number of nurses for shift s ($1 \leq s \leq S$) at day d ($1 \leq d \leq D$).

Let $\mathbf{N} = \{N_1, N_2, \ldots, N_n\}$ define the set of n available nurses. Moreover, let $\mathbf{M} \subseteq \mathbf{N}$ define the subset of nurses with managerial qualifications.

Nurses are to be assigned to work shifts. However, the daily work schedule for a nurse must be continuous, i.e., a nurse cannot be assigned a morning shift and a night shift in the same day with a pause in the middle. Moreover, each nurse N_k has a minimum and maximum number of daily workload. Let L_k^{min} and L_k^{max} define the minimum and maximum number of shifts for

Shift	Day 1	2	3
1	2	2	2
2	3	3	1
3	2	3	3
4	3	2	1

Table 1: Shift's requirements.

Nurses	Manager	L_k^{min}	L_k^{max}	H_k
N_1	Y	1	3	7
N_2	Y	2	3	7
N_3	N	1	3	4
N_4	N	2	2	6
N_5	N	1	2	5

Table 2: Nurse's constraints.

Shift	Day 1	2	3
1	[1, -1, 3, 1, 3]	[-1, 2, 4, -1, -1]	[1, -1, -1, 1, -1]
2	[2, 1, 1, 1, 2]	[2, 2, 1, 4, 1]	[2, 1, 1, 1, 2]
3	[-1, 1, 1, 3, 1]	[1, 2, -1, 1, 1]	[1, 1, 1, 1, 3]
4	[1, 1, 2, -1, 1]	[1, -1, -1, 2, 3]	[1, 1, 3, 3, 1]

Table 3: Nurse's Preferences for each shift in the period.

nurse N_k ($1 \leq k \leq n$) in any given day. Additionally, let H_k define the maximum number of shifts that can be assigned to nurse N_k during the number of days D being considered in the schedule.

Finally, each nurse defines its preference for each shift s and day d. Let $P_{k,s,d}$ define the cost of assigning nurse N_k to shift s in day d. Hence, shifts with lower costs are preferred to the ones with higher costs. The cost of an available shift is always positive. Whenever we have a value of -1 to $P_{k,s,d}$, it means that nurse N_k is not available in day d for shift s.

Your goal is to help the healthcare facility to define the nurse's schedule for D days such that:

- Each shift s of day d has at least $R_{s,d}$ nurses assigned;
- Each shift s of day d has at least one nurse with managerial qualifications;
- The workload of a nurse in a given day is continuous;
- During a day, the number of shifts for a nurse N_k is always between L_k^{min} and L_k^{max};
- A nurse N_k is never assigned more than H_k shifts in the period;
- A nurse N_k is never assigned to a shift where it is declared as unavailable;
- The total cost of assigning nurses to shifts in minimized;

Shift	Day 1	2	3
1	$[N_1, N_4]$	$[N_2, N_3]$	$[N_1, N_4]$
2	$[N_1, N_2, N_4]$	$[N_2, N_3, N_5]$	$[N_1, N_4]$
3	$[N_2, N_5]$	$[N_1, N_4, N_5]$	$[N_1, N_2, N_3]$
4	$[N_2, N_3, N_5]$	$[N_1, N_4]$	$[N_2, N_5]$

Table 4: Optimal solution for the example data in Tables 1, 2 and 3.

Example

Table 1 provides an example of the minimal number of nurses necessary for a schedule with 3 days and 4 shifts per day. Additionally, consider Table 2 with the data for 5 nurses. Table 2 defines for each nurse N_k if N_k can be the manager of a shift as well as the values of L_k^{min}, L_k^{max} and H_k. Finally, Table 3 shows the preferences for each nurse in each shift of the period.

Table 4 shows an optimal solution for this example. The overall cost of this solution is 38. Note that all constraints of the problem are satisfied. Moreover, observe that shift 2 in day 3 has more nurses than the minimum due to the constraints on the minimum number of consecutive shifts of N_4.

Project Details

You are to implement a tool, or optionally a set of tools, invoked with command `proj1`. Your implementation must use a SMT tool to solve the Nurse Scheduling Problem as previously specified.

Your tool does not take any command-line arguments. The problem instance is to be read from the standard input.

Consider an instance file named `instance.nsp`. The tool is expected to be executed as follows:

```
proj2 < instance.nsp > solution.out
```

The tool must write the solution to the standard output, which can then be redirected to a file (e.g., `solution.out`).

The programming languages to be used are only Python, C/C++ or Java. The formats of the files used by the tool are described below.

File Formats

You can assume that all input files follow the description provided in this document. There is no need to check if the input is correct. Additionally, all lines (input or output) must terminate with the end-of-line character.

Input Format

The input file representing a problem instance is a text file that follows the following format:

- One line with two integers D and S defining the number of days (D) and the number of shifts per day (S);
- A sequence of D lines where each line contains a sequence of S positive integers; The ith integer of line j defines the minimum number of nurses of shift i in day j;
- One line with an integer defining the number of nurses n;
- A sequence of n lines where each line defines the nurse's constraints; The kth line of the sequence defines the constraint of nurse N_k as follows:
  - A character (Y/N) defining if nurse has managerial qualifications;
  - Three integers defining L_k^{min}, L_k^{min} and H_k.
- A sequence of $D * S$ lines where each line defines the nurses preferences of a shift; The first S lines defines the preferences of day 1, the next S lines the preferences for day 2, etc. Each of the $D * S$ lines contains a sequence of n integers where the kth integer defines the preference of nurse N_k for that shift (-1 to denote unavailability);

There is always one white space to separate the data in the same line. Moreover, all lines end with an end-of-line character.

Output Format

The output representing an optimal solution to the problem instance must comply with the following format:

- One line with an integer defining the overall optimal value of nurse's preferences;
- A sequence of $D * S$ lines defining the nurses assigned to each shift; The first S lines define the assignments for day 1, the next S lines define the shifts for day 2, etc. Each line must contain:
  - An integer m defining the total number of nurses assigned to the shift;
  - A sequence of m integers where each integer is a value between 1 and n representing the ID of the shift's nurses;

Important: The final version to be submitted for evaluation must comply with the described output. Project submissions that do not comply will be severely penalized, since each incorrect output will be considered as a wrong answer. An application that verifies if the output complies with the description will be available on the course's website.

Example 1

```
3 4
2 3 2 3
2 3 3 2
2 1 3 1
5
Y 1 3 7
Y 2 3 7
N 1 3 4
N 2 2 6
N 1 2 5
1 -1 3 1 3
2 1 1 1 2
-1 1 1 3 1
1 1 2 -1 1
-1 2 4 -1 -1
2 2 1 4 1
1 2 -1 1 1
1 -1 -1 2 3
1 -1 -1 1 -1
2 1 1 1 2
1 1 1 1 3
1 1 3 3 1
```

An optimal solution would be:

```
38
2 1 4
3 1 2 4
2 2 5
3 2 3 5
2 2 3
3 2 3 5
3 1 4 5
```

```
2 1 4
2 1 4
2 1 4
3 1 2 3
2 2 5
```

Additional Information

The project is to be implemented in groups of one or two students.

The project is to be submitted through the git repository of the group. The faculty will create the group's repositories with a sample of problem instances. Jointly with your code, the repository must also contain a short report describing the main features of your project.

The report to be submitted must be in **pdf** format and should describe: (1) the problem to be solved, (2) how to install and run your project, (3) a description of the encodings used to solve the problem (i.e., the sets of variables and constraints used in your formula), (4) the algorithm and configurations used to solve it and (5) experimental results showing the differences in performance of the different encodings implemented.

The evaluation will be made taking into account correctness and efficiency given a reasonable amount of CPU time (80%) and the report (20%).

The input and output formats described in this document must be strictly followed.

Project Dates

- Project published: 07/10/2025.
- Project due: 20/10/2025 at 23:59.

Omissions & Errors

Any detected omissions or errors will be added to future versions of this document. Any required clarifications will be made available through the course's official website.

Versions

07/10/2025, version 1.0: Original version.